

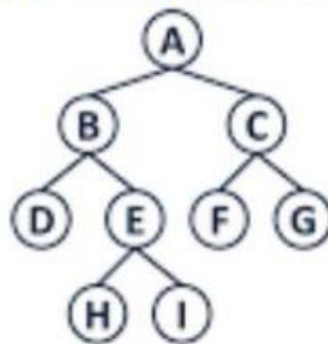
1. Write the C function to add two polynomials. Show the linked representation of the below two polynomials and their addition using a circular singly linked list

P1: $5x^3 + 4x^2 + 7x + 3$

P2: $6x^2 + 5$

Output: add the above two polynomials and represent them using the linked list.

2. Write recursive C functions for inorder, preorder and postorder traversals of a binary tree. Also, find all the traversals for the given tree.



or

Construct a binary tree from the Post-order and In-order sequence given below

In-order: GDHBAEICF

Post-order: GHDBIEFCA

Or

Define binary tree with an example. Write a C recursive routine to traverse the given tree using inorder, preorder and postorder

3. Define Sparse matrix. For the given sparse matrix, give the linked list representation:

$$A = \begin{bmatrix} 0 & 0 & 3 & 0 & 4 \\ 0 & 0 & 5 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 6 & 0 & 0 \end{bmatrix}$$

4. Define the Threaded binary tree. Construct Threaded binary for the following elements: A, B, C, D, E, F, G, H, I

Or

What is the advantage of the threaded binary tree over binary tree? Explain the construction of threaded binary tree for 10 20 30 40 and 50

5. Define Binary Search tree. Construct a binary search tree (BST) for the following elements: 100, 85, 45, 55, 120, 20, 70, 90, 115, 65, 130, 145.

Traverse using in-order, pre-order, and post order

Or Define binary search tree. Draw the BST the following input: 14 15 4 9 7 18 3 5
16 20 17 9